

Figure 27.—Transmutation of lithium nucleus into two helium nuclei by a proton (*after Kirchner*).

In other words, two alpha particles, or helium nuclei, are formed. Figure 27 (after a photograph by Kirchner) shows this process taking place in the cloud chamber. The end of the discharge tube, in which the protons are accelerated, is visible. From there they impinge on a piece of metallic lithium. We can see the tracks of two alpha particles start there and proceed in opposite directions. (The other visible track does not belong to this reaction.) A similar reaction is shown in Figure 28. Here a boron nucleus changes, in consequence of the absorption of a fast proton, into three helium nuclei, i.e.:

$$_5B^{11} + _1H^1 \rightarrow {}_2He^4 + {}_2He^4 + {}_2He^4$$